

Quick IPT — Hidden-Point Measurement for the EVKA Position System

EVKA Position Project

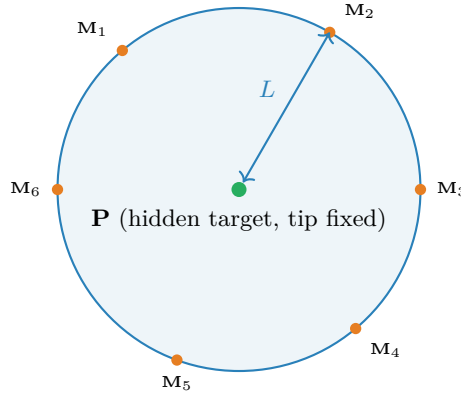
July 8, 2026

1 What is Quick IPT?

Quick IPT (Inverted Pen Technology) lets you **measure a point you cannot touch directly** — behind an obstacle, around a corner, inside a recess. You hold the pen *tip* fixed on the hidden target and sweep the *handle* in a wide spiral. The wire attaches to the pen body at a fixed offset L from the tip, so every point the sensor measures lies on a sphere of radius L centred on the target. Fitting that sphere recovers the target.

The method was pioneered by **Prodim International** in the **Proliner V9** handheld 3D scanner (US Patent 9,267,794 B2). The Proliner V9 uses the same sensor topology as `evka_position`: one draw-wire encoder for length and two rotary encoders for angles. This means the method transfers directly — **no firmware change is needed**; the entire feature is a host-side Python tool that reads the live position stream and does the math.

All measured points $\mathbf{M}_i$ lie on a sphere of radius L centred on $\mathbf{P}$



The core equation is simply:

$$\|\mathbf{M}_i - \mathbf{P}\| = L \quad \text{for every sample } i \quad (1)$$

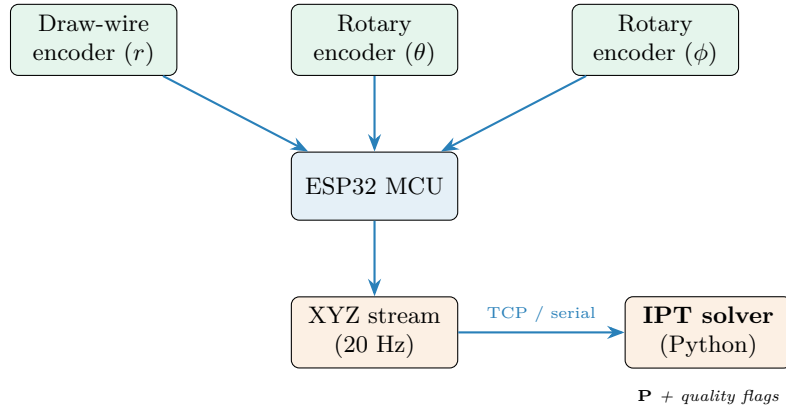
The solver fits the sphere to the measured point cloud and returns the centre $\mathbf{P}$ — the hidden target. It also returns the fitted radius L_{hat} , which *self-calibrates* the pen offset: you never have to measure L precisely.

2 Proliner V9 and the same idea

The **Prodim Proliner V9** is a portable 3D scanner for the stone, glass, and metal fabrication industry. The operator touches points with a pen connected by a wire to a base unit. The

Proliner’s *Quick IPT* feature extends this to hidden points: the pen tip stays fixed on the target while the handle sweeps, and the device computes the sphere centre.

EVKA Position has the same sensor topology:



The Proliner V9 and EVKA Position both produce 3-DOF points (Cartesian XYZ), not 6-DOF poses. This is why multiple samples and a wide sweep are needed: the sphere fit recovers the centre from the point cloud alone, without any orientation information.

3 How it works

3.1 The measurement

1. **Connect** the Python tool to the EVKA device over TCP or serial.
2. Hold the pen **tip** firmly on the hidden target.
3. Click **ARM** and sweep the handle in a **wide spiral**.
4. Click **STOP**, then **SOLVE**.



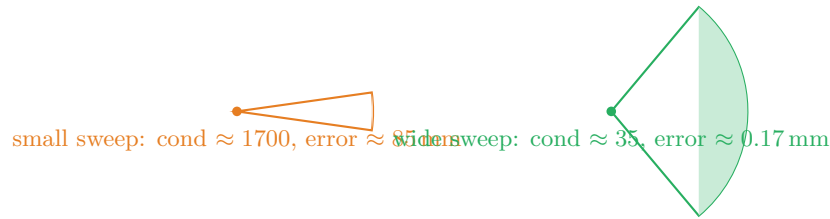
3.2 Two solver modes

- **Self-calibrating** (leave L blank): fits both centre and radius. L_{hat} is the fitted pen length — should match your physical pen within a few mm.
- **Known L** (enter pen length): uses a better-conditioned trilateration method. L_{used} is the value you entered; L_{fit} is the independent sphere-fit radius — if they differ by more than a few mm, the entered length is wrong or the tip slipped.

3.3 Quality flags

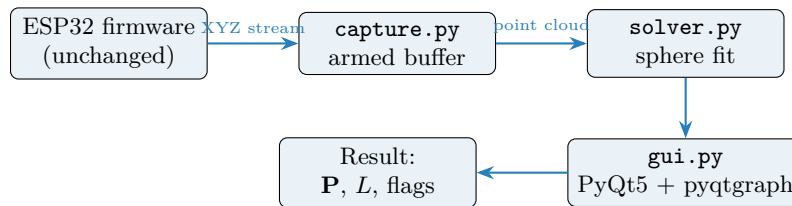
Flag	Metric	Bands
slip_warning	RMS residual	ok < 2 mm, warn 2–5 mm, reject > 5 mm
geom_warning	Geometric condition	ok < 150, warn 150–1500, block > 1500

RMS residual tells you if the tip slipped. **Geometry cond** tells you if the sweep was too small. Both are heuristic thresholds tuned in simulation — re-tune on real hardware.



A small patch of a large sphere is nearly flat — it cannot pin the centre. This is geometric dilution of precision, exactly like poor satellite geometry in GNSS. **Sweep wider.**

4 Implementation



The code lives in `tools/ipt/` — a small Python package:

```

1 tools/ipt/
2 __main__.py    # CLI: python -m tools.ipt --tcp 192.168.1.50:8080
3 solver.py      # Sphere fit + trilateration + Gauss-Newton (pure numpy)
4 capture.py     # Armed buffer: dedup, validity, finite-value guard
5 gui.py         # PyQt5 window: 3D cloud, fitted sphere, controls
6 tests/         # Offline pytest suite (no hardware needed)
  
```

No firmware upload. The tool connects to the existing EVKA device and reads the live 20 Hz position stream. The ESP32 code is unchanged.

Transport asymmetry

The firmware sends different line formats over serial vs. TCP:

- **Serial:** one `DATA,...` line per cycle (XYZ + validity).
- **TCP:** separate `X,Y,Z` and `SENSOR,...` lines. The capture buffer pairs them to recover validity.

At 20 Hz the firmware emits many near-identical frames when the handle is still. The buffer deduplicates points closer than 1 mm to the last accepted sample.

5 How to use it

```

1 # WiFi (connect to CMD CNC_EVKA AP at 192.168.1.50)
2 python -m tools.ipt --tcp 192.168.1.50:8080
3
4 # Serial
5 python -m tools.ipt --serial /dev/ttyUSB0 --baud 115200
6
7 # No flag -- opens disconnected, pick transport in the panel
8 python -m tools.ipt
  
```

Then: connect, hold tip on target, ARM, sweep wide, STOP, SOLVE.

The recovered point **P** is in the **sensor origin frame** (the boot zero-point), in mm — the same frame as the live X/Y/Z readout, not a world frame.

6 Limitations

- **3-DOF only.** No orientation sensor; IPT recovers the hidden *point*, not a 6-DOF pose.
- **Tip slip.** RMS residual flags it but cannot fix it. Re-measure.
- **Sweep geometry.** Small sweeps give poor conditioning. The `geom_warning` flag is your cue to sweep wider.
- **Coordinate frame.** Results are relative to the sensor’s boot zero-point. If the sensor is moved or re-zeroed, the recovered target is no longer valid.

Intellectual property note

The Quick IPT method is patented by Prodim International (US 9,267,794 B2, EP 2,792,994 B1). This implementation is a technical reverse-engineering exercise for research and educational purposes. The mathematical methods (sphere fitting, pivot calibration) are well-established in the literature. Commercial deployment may require licensing from Prodim.

References

- [1] Z. Yaniv, “Which Pivot Calibration?”, *SPIE Medical Imaging*, vol. 9415, 2015.
- [2] I. D. Coope, “Circle fitting by linear and nonlinear least squares”, *J. Opt. Theory Appl.*, vol. 76, no. 2, pp. 381–388, 1993.
- [3] E. Teune and J. Janssen, “Method and device for determining a position of a measurement tip”, US Patent 9,267,794 B2, 2016.